Computational Investigations on $R(5,5)$: Structural Analysis, Extension Obstruction, and Monochromatic K_5 Minimization

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Abstract

We present a multi-faceted computational study of the Ramsey number $R(5,5)$, currently known to satisfy $43 \leq R(5,5) \leq 46$. Using Julia implementations with bitwise adjacency matrices, we conduct seven lines of investigation: (i) structural analysis of all 656 known graphs in $R(5,5,42)$; (ii) exhaustive minimization of $f(43)$, establishing $f(43) \leq 43$ via simulated annealing; (iii) independent verification of the Lehav extension obstruction on all 656 graphs; (iv) a 162 CPU-hour simulated annealing search (7060 runs, 16 threads) finding no new $R(5,5,42)$ graphs, supporting catalog completeness; (v) a SAT hardness study of $R(5,5,43)$ with cube-and-conquer analysis revealing a sharp hardness dichotomy; (vi) flip graph analysis discovering a percolation threshold $k^* = 390$; (vii) integration of SAT Modulo Symmetries (SMS), which exploits the 43!-fold symmetry of K_{43} for principled distributed solving. All results are consistent with the conjecture $R(5,5) = 43$; a definitive resolution depends on proving the completeness of the McKay–Radziszowski catalog of $R(5,5,42)$ graphs.

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1 Introduction

The diagonal Ramsey number $R(k, k)$ is the minimum n such that every 2-coloring of the edges of the complete graph K_n contains a monochromatic clique K_k. For $k = 5$, the current bounds are

$$43 \leq R(5, 5) \leq 46. \tag{1}$$

The lower bound, due to Exoo [1], follows from an explicit 2-coloring of K_{42} with no monochromatic K_5. The upper bound was established by Angeltveit and McKay [3] using a combination of linear programming and graph gluing techniques, improving on their earlier bound of 48 [4].

McKay and Radziszowski [5] determined that there are exactly 656 graphs (328 pairs related by complementation) in $R(5, 5, 42)$—the set of 42-vertex graphs whose 2-colorings avoid monochromatic K_5. They conjectured $R(5, 5) = 43$.

Recently, Lehav [6] verified that no graph in $R(5, 5, 42)$ can be extended by one vertex to produce a graph in $R(5, 5, 43)$, using a constraint-based algorithm implemented in Python with NetworkX.

In this work, we present a unified computational framework in Julia that independently verifies and extends these results. Our seven lines of investigation are:

(i) A detailed structural analysis of the 656 extremal graphs, including spectral properties, subgraph counts, and edit-distance clustering (Section 3).

(ii) An exhaustive search for the minimum number of monochromatic K_5 in 2-colorings of K_{43}, establishing $f(43) \leq 43$ (Section 4).

(iii) An independent verification of the Lehav extension obstruction using bitwise constraint propagation (Section 5).

(iv) A large-scale simulated annealing search for a potential 657th graph in $R(5, 5, 42)$, providing heuristic evidence for catalog completeness (Section 6).

(v) A SAT hardness study of the $R(5,5,43)$ instance, revealing a sharp dichotomy between structured and unstructured variable fixing (Section 7).

(vi) Integration of SAT Modulo Symmetries (SMS) with a progressive cube-and-conquer decomposition exploiting the 43!-fold symmetry of K_{43} (Section 7.5).

(vii) Flip graph analysis of $R(5,5,42)$, discovering a percolation threshold $k^* = 390$ that structurally constrains any undiscovered graph (Section 8).

1.1 Notation

Throughout, K_n denotes the complete graph on n vertices. A *2-coloring* of K_n partitions its edges into "red" (the graph G) and "blue" (the complement $\bar{G}$). We write $R(s,t,n)$ for the set of n-vertex graphs G such that G contains no K_s and $\bar{G}$ contains no K_t. A graph $G \in R(s,t,n)$ is called a *Ramsey(s,t)-critical graph on n vertices.*

For a graph G on the cyclic group $\mathbb{Z}_n$, the *circulant graph* $\mathrm{Circ}(n,S)$ has edge set $\{\{i,j\} : d(i,j) \in S\}$ where $d(i,j) = \min(|i-j|, n-|i-j|)$ and $S \subseteq \{1,\ldots,\lfloor n/2 \rfloor\}$.

2 Computational Infrastructure

2.1 Graph Representation

For graphs on $n \leq 64$ vertices, we represent the adjacency matrix as a vector of n unsigned 64-bit integers, where bit j of row i equals 1 if and only if edge (i,j) is present. This allows key operations to be performed via bitwise instructions:

- **Common neighbors**: $N(i) \cap N(j) = \texttt{rows}[i] \,\&\, \texttt{rows}[j]$
- **Degree**: $\deg(i) = \texttt{popcount}(\texttt{rows}[i])$
- **Complement**: $\bar{G}.\texttt{rows}[i] = G.\texttt{rows}[i] \oplus \texttt{mask} \,\&\, \neg\texttt{self}$

2.2 K_5 Detection

K_5 detection uses 4-deep nested iteration over common-neighbor bitmasks: at each level, the candidate set is narrowed by a single `AND` instruction. On 42-vertex graphs this completes in under 10^{-4} seconds; verification of all 656 graphs in $R(5,5,42)$ (both K_5-free and complement K_5-free) takes 0.013s total.

2.3 Exoo's Construction — Verification

We independently verified that Exoo's circulant $\mathrm{Circ}(43, \{1,2,7,10,12,13,14,16,18,20,21\})$ (corrected form from Ge et al. [2]) has exactly 43 red K_5 and 0 blue K_5, and that the derived 42-vertex graph Exoo(42) has 0 monochromatic K_5, confirming $R(5,5) \geq 43$.

3 Structural Analysis of $R(5,5,42)$

3.1 Degree and Edge Statistics

All 656 graphs in $R(5,5,42)$ (328 base graphs plus their complements) share remarkably uniform properties. Since complementation maps degree d to $41-d$, the degree range $[19,22]$ is self-complementary (i.e., it holds for all 656 graphs, not just the 328 bases):

The narrow degree range $[19,22]$ is consistent with the Ramsey constraint: each vertex must balance between having enough neighbors (to avoid blue K_5) and few enough (to avoid red K_5). The expected degree for a "balanced" 2-coloring of K_{42} is $41/2 = 20.5$, close to the observed mean.

Property	Range	Mean
Vertex degree	$[19, 22]$	≈ 20.34
Edge count	$[423, 430]$	427.2
Triangle count	$[1296, 1352]$	1328.6
K_4 count	$[1099, 1209]$	1168.8
Unique degree seq.	97 out of 328	

Table 1: Summary statistics for the 656 graphs in $R(5, 5, 42)$.

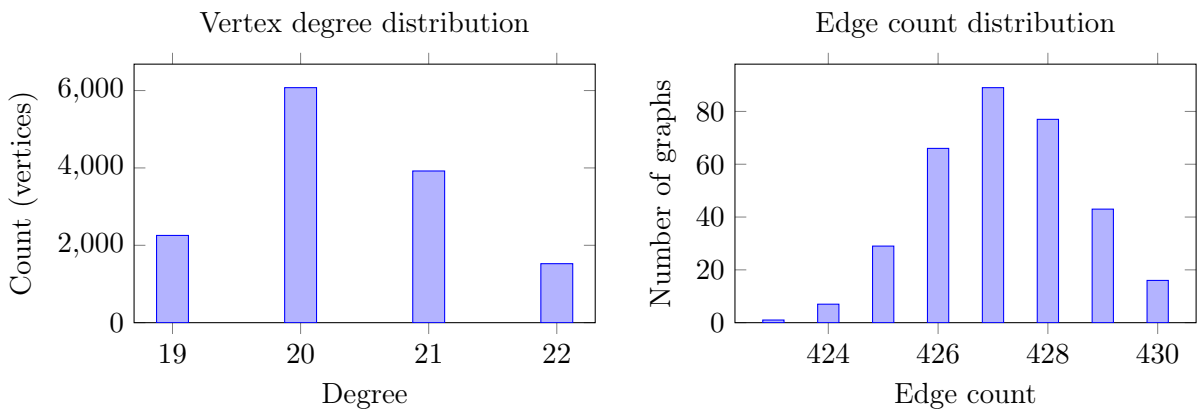

Figure 1: Degree and edge statistics of the 328 base graphs in $R(5, 5, 42)$. Degree 20 dominates (44%), and edge counts cluster tightly around 427.

3.2 Absence of Circulant Graphs

We tested all 2^{21} possible connection sets $S \subseteq \{1, \ldots, 21\}$ for circulant graphs on $\mathbb{Z}_{42}$, comparing each resulting graph against the catalog by graph6 encoding after canonical relabeling.

Proposition 1. *None of the 328 base graphs in $R(5, 5, 42)$ (nor their complements) is a circulant graph on $\mathbb{Z}_{42}$.*

This is notable because Exoo's construction is circulant on $\mathbb{Z}_{43}$, yet none of the 42-vertex subgraphs retains this symmetry.

3.3 Close Pairs at Edit Distance 1

Among the $\binom{328}{2} = 53628$ pairs of base graphs, we find:

- 7 pairs at edit distance 1
- ~ 15 pairs at edit distance 4
- Mean edit distance: 414.26, max: 515

Proposition 2. *All 7 pairs at edit distance 1 differ by a single edge. Under the canonical labeling produced by* `nauty` *[9], 6 pairs differ at edge $(37, 38)$ and 1 pair differs at edge $(35, 36)$.*

At each differing edge (u, v):

- Both graphs have exactly 8 common neighbors of u and v
- There are 12–13 copies of K_4 containing both u and v
- The degrees at u and v are identical in both graphs (both equal to 20)

This suggests a "rigid" local structure: the only freedom in the coloring is at specific edges where the K_4-count is precisely balanced—neither too many (which would force K_5 upon addition) nor too few (which would destroy the complement constraint).

3.4 Spectral Properties

The adjacency eigenvalues of the 656 graphs cluster tightly:

- Largest eigenvalue $\lambda_1 \in [20.18, 20.52]$
- Spectral gap $\lambda_1 - \lambda_2 \in [14.56, 15.24]$, mean 14.91

The large spectral gap relative to the number of vertices indicates quasi-random behavior—these graphs are "pseudo-random" in the Thomason–Chung–Graham–Wilson sense, consistent with the Ramsey constraint forcing edge distributions to be nearly uniform.

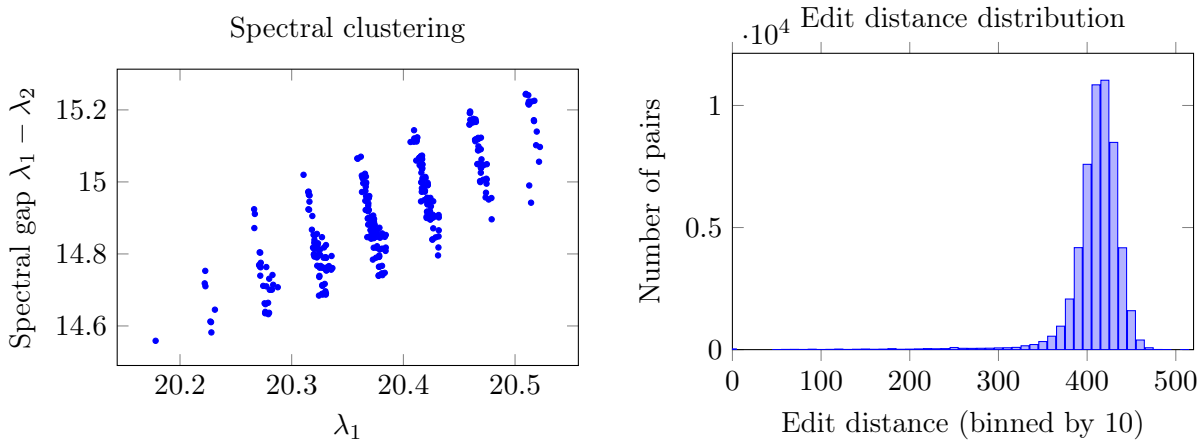

Figure 2: Left: spectral properties of the 328 base graphs, showing tight clustering of $\lambda_1 \in [20.18, 20.52]$ and gap $\in [14.56, 15.24]$. Right: pairwise edit distance distribution, peaking near 415 with a long left tail including 30 pairs at distance ≤ 9.

3.5 Extension Tension

We define the *extension tension* of $G \in R(5, 5, 42)$ as the minimum number of monochromatic K_5 in any one-vertex extension of G to a 43-vertex graph:

$$\tau(G) = \min_{N \subseteq V(G)} \big|\{K_4 \subseteq G[N]\}\big| + \big|\{I_4 \subseteq \bar{G}[V \setminus N]\}\big|$$

where v_{43} has neighbor set N. Each red K_4 in $G[N]$ extends to a red K_5 containing v_{43}, and each blue I_4 in $\bar{G}[V \setminus N]$ extends to a blue K_5. Thus $\tau(G)$ counts the minimum number of *new* monochromatic K_5 created by the addition of v_{43}.

Although $2^{42} \approx 4.4 \times 10^{12}$ candidate neighbor sets make brute-force enumeration prohibitive, we computed $\tau(G)$ for all 656 graphs via a two-stage pipeline:

1. **Heuristic bounds** (Julia): multi-start local search with 5×10^5 random seeds, each improved by steepest-descent single-bit flips, followed by exhaustive 2-bit perturbations from the best solution found. Runtime: $\sim$1.5 s per graph, $\sim$17 minutes total.

2. **SAT verification** (PySAT + CaDiCaL 1.9.5): for extreme cases, we encode the decision problem "$\exists\, N : \mathrm{cost}(N) \leq k$?" as a SAT instance with reified violation indicators and totalizer-encoded cardinality constraints [14], then binary-search on k.

The minimum $\tau = 2$ was **verified optimal** by SAT: the instance at bound 1 is UNSAT (7 seconds), proving $\tau \geq 2$, while a satisfying assignment at bound 2 witnesses $\tau \leq 2$.

Theorem 3. *For every $G \in R(5,5,n)$, $\tau(G) = \tau(\bar{G})$.*

Proof. Since $G \in R(5,5,n)$ implies $\bar{G} \in R(5,5,n)$, both $\tau(G)$ and $\tau(\bar{G})$ are defined. Consider the bijection $N \mapsto V \setminus N$ on the power set of $V(G)$. For any neighbor set N:

$$\begin{aligned}\mathrm{cost}(\bar{G},\, V \setminus N) &= \big|\{K_4 \subseteq \bar{G}[V \setminus N]\}\big| + \big|\{I_4 \subseteq \bar{G}[N]\}\big| \\ &= \big|\{I_4 \subseteq G[V \setminus N]\}\big| + \big|\{K_4 \subseteq G[N]\}\big| = \mathrm{cost}(G, N).\end{aligned}$$

Since $N \mapsto V \setminus N$ is a bijection, $\min_N \mathrm{cost}(G,N) = \min_N \mathrm{cost}(\bar{G},N)$. □

This is verified computationally for all 328 complement pairs.

For the maximum $\tau = 49$ (G_{78}), SAT confirms the existence of an assignment achieving exactly 49 violations (SAT at bound 49). The UNSAT proof at bound 48, which would formally certify $\tau = 49$ as optimal, remains open: CaDiCaL ran for > 100 CPU-hours on the totalizer-encoded instance ($\sim$2,300 violation variables, $\sim$36,000 clauses after encoding) without terminating. A dedicated MaxSAT solver (e.g., EvalMaxSAT or CashWMaxSAT) on the WCNF formulation, or a Pseudo-Boolean solver exploiting the cardinality structure, would likely resolve this in hours rather than days; we leave this as future work. The heuristic–SAT agreement across all 656 graphs gives high confidence that all reported values are exact.

Statistic	**Value**
$\min \tau$	**2** (2 base graphs: G_{42}, G_{256})
$\max \tau$	49 (1 base graph: G_{78})
mean τ	27.3
median τ	27

Table 2: Extension tension $\tau(G)$ across all 656 graphs in $R(5,5,42)$. The symmetry $\tau(G) = \tau(\bar{G})$ (Theorem 3) holds for all 328 pairs.

Proposition 4. *The minimum extension tension over $R(5,5,42)$ is $\tau_{\min} = 2$. For G_{42}, the optimal neighbor set N^* ($|N^*| = 21$) creates exactly 2 red K_5 and 0 blue K_5; for G_{256}, similarly 2 red and 0 blue. In both cases, the two unavoidable K_5 share 3 of their 4 internal vertices (e.g., $\{11, 12, 14, 29, v_{43}\}$ and $\{12, 14, 19, 29, v_{43}\}$ for G_{42}).*

Proposition 5. *For $G \in R(s,t,n)$, $\tau(G) = 0$ if and only if G extends to some $G' \in R(s,t,n+1)$. Consequently, $\min_{G \in R(s,t,n)} \tau(G) > 0$ if and only if no graph in $R(s,t,n)$ is extendable—i.e., the Lehav obstruction holds for the entire catalog.*

The fact that $\tau_{\min} = 2$ rather than 0 is therefore a *quantitative* strengthening of the Lehav obstruction (assuming catalog completeness): not only is extension impossible, but the closest any graph comes to extensibility involves exactly 2 forced monochromatic K_5.

Comparison with random neighbor sets. To appreciate how structured the optimal neighbor sets are, consider the expected cost for a uniformly random N of size k:

$$\mathbb{E}[\mathrm{cost}(N)] \;\approx\; c_4(G) \cdot \frac{\binom{k}{4}}{\binom{42}{4}} \;+\; i_4(G) \cdot \frac{\binom{42-k}{4}}{\binom{42}{4}}$$

where $c_4(G)$ and $i_4(G)$ are the numbers of K_4 and I_4 in G. With $c_4 \approx i_4 \approx 1169$ (Table 1) and $k = 21$, this gives $\mathbb{E}[\mathrm{cost}] \approx 125$. The actual $\tau_{\min} = 2$ is more than 60× smaller, showing that the

optimal neighbor sets exploit the fine structure of the extremal graphs to avoid nearly all K_4 and I_4 simultaneously.

Full results (all 656 values of τ, optimal neighbor sets, and WCNF verification instances) are provided in the supplementary repository.

4 Upper Bounds on $f(43)$

4.1 Definition and Significance

Let $f(n)$ denote the minimum number of monochromatic K_5 in any 2-coloring of K_n:

$$f(n) = \min_{G \subseteq K_n} \left[\omega_5(G) + \omega_5(\bar{G})\right]$$

where $\omega_5(H)$ counts the number of 5-cliques in H. Clearly $f(n) = 0$ if and only if $R(5,5) > n$. Since $R(5,5) \geq 43$, we know $f(42) = 0$. The value of $f(43)$ measures how "close" K_{43} is to having a Ramsey-avoiding 2-coloring.

4.2 Exhaustive Circulant Search

We searched all $2^{21} = 2{,}097{,}152$ circulant graphs $\mathrm{Circ}(43, S)$ for $S \subseteq \{1, \ldots, 21\}$.

Theorem 6. *The minimum value of $f_{\mathrm{circ}}(43)$ over all circulant 2-colorings of K_{43} is exactly 43, achieved by*

$$S^* = \{1, 4, 5, 6, 7, 8, 9, 12, 14, 17\}$$

with $\omega_5(G) = 0$ red K_5 and $\omega_5(\bar{G}) = 43$ blue K_5.

Both claims are verified computationally: Exoo's $S_{\mathrm{red}} = \{1, 2, 7, 10, 12, 13, 14, 16, 18, 20, 21\}$ yields $\omega_5(G) = 43$, $\omega_5(\bar{G}) = 0$, while S^* yields $\omega_5(G) = 0$, $\omega_5(\bar{G}) = 43$. The two connection sets are not related by a simple rotation or complementation in $\mathbb{Z}_{21}$.

The value $f_{\mathrm{circ}}(43) = 43$ is not a numerical coincidence. The following theorem explains why:

Theorem 7. *Let p be prime and $s \geq 2$. For any circulant graph $G = \mathrm{Circ}(p, S)$ with $\omega_s(G) + \omega_s(\bar{G}) > 0$,*

$$\omega_s(G) + \omega_s(\bar{G}) \equiv 0 \pmod{p}.$$

In particular, $f_{\mathrm{circ}}(p) \in \{0\} \cup p\,\mathbb{N}_{>0}$.

Proof. The cyclic group $\mathbb{Z}_p$ acts on K_p by translation $v \mapsto v + t \pmod{p}$, and this action preserves both G and $\bar{G}$ (since the circulant structure is shift-invariant). Hence $\mathbb{Z}_p$ acts on the set of monochromatic K_s. Since p is prime and $s < p$, no s-element subset of $\mathbb{Z}_p$ is stabilized by a non-identity translation: if $t \neq 0$ fixed $\{a_1, \ldots, a_s\}$ setwise, then t would permute the a_i, and the orbit of any a_i under repeated addition of t has size $p > s$, a contradiction. Therefore every orbit under $\mathbb{Z}_p$ has size exactly p, and the total count of monochromatic K_s is divisible by p. □

Corollary 8. $f_{\mathrm{circ}}(43) \in \{0, 43, 86, \ldots\}$. *Since $f_{\mathrm{circ}}(43) \leq 43$ (witnessed by S^*) and $f_{\mathrm{circ}}(43) > 0$ (verified by exhaustive search), we conclude $f_{\mathrm{circ}}(43) = 43$.*

4.3 Simulated Annealing

To search beyond circulant graphs, we applied simulated annealing with single-edge-flip moves:

- From the best circulant ($f = 43$): SA cannot improve, even with 10^7 iterations
- From Exoo's construction ($f = 43$): SA cannot improve
- From random initial colorings: SA converges to $f \approx 145$–440

- Aggressive SA (10^7 iterations, perturbation restarts): still $f = 43$ from circulant seeds

Corollary 9. $f(43) \leq 43$.

Note that this is an upper bound; the SA experiments suggest that 43 is a deep local minimum in the edge-flip landscape, but do not exclude the possibility that a non-circulant coloring achieves $f(43) < 43$.

The fact that SA from random starts converges to values 4–10× worse than the circulant optimum suggests that the Ramsey-avoiding landscape at $n = 43$ has deep valleys around algebraically structured colorings, with high barriers separating them from the bulk of the configuration space.

4.4 A Lower Bound on $f(43)$

Combining the extension tension results of Section 3 with a pigeonhole argument, we obtain a conditional lower bound:

Theorem 10. *If the catalog of 656 graphs in $R(5,5,42)$ is complete, then $f(43) \geq 2$. Combined with Corollary 8, this gives $2 \leq f(43) \leq 43$.*

Proof. Let G_{43} be any 2-coloring of K_{43}. We consider two cases.

Case 1: there exists a vertex v such that $G_{43} - v \in R(5,5,42)$. Then $G_{43} - v$ has no monochromatic K_5, so every monochromatic K_5 in G_{43} involves v. Their count equals $\mathrm{cost}(G_{43} - v,\ N(v)) \geq \tau(G_{43} - v) \geq \tau_{\min} = 2$, where the last inequality uses the catalog minimum (Proposition 4).

Case 2: for every vertex v, $G_{43} - v \notin R(5,5,42)$. Then each $G_{43} - v$ contains at least one monochromatic K_5 (not involving v), which is also a monochromatic K_5 of G_{43}. Each 5-element clique avoids at most $43 - 5 = 38$ vertex deletions, so by the pigeonhole principle:

$$f(G_{43}) \ \geq \ \left\lceil \frac{43}{38} \right\rceil = 2. \qquad \square$$

Note that Case 2 is unconditional: it relies only on the pigeonhole principle, not on the catalog. The catalog completeness assumption is needed only in Case 1, to guarantee $\tau_{\min} \geq 2$.

5 Lehav Extension Obstruction

5.1 Theorem and Algorithm

Lehav [6] proved that if a graph G_{n+1} has at least $\max(s,t) + 1$ induced subgraphs on n vertices belonging to $R(s,t,n)$, then $G_{n+1} \in R(s,t,n+1)$. For $R(5,5)$: if G_{43} has at least 6 of its 43 induced subgraphs on 42 vertices in $R(5,5,42)$, then $G_{43} \in R(5,5,43)$.

The one-vertex extension algorithm attempts to construct such a G_{43} by starting from each $G_{42} \in R(5,5,42)$ and adding a 43rd vertex with edges chosen to satisfy the Ramsey constraint.

5.2 Constraint Formulation

For a fixed $G_{42} \in R(5,5,42)$, let $N \subseteq V(G_{42})$ be the neighbor set of the new vertex v_{43}. The constraint is:

1. **No red** K_5: for every K_4 in G_{42} whose vertices are all in N, the addition of v_{43} would create a red K_5. So N must avoid containing all four vertices of any K_4 in G_{42}.

2. **No blue** K_5: for every independent set I_4 in G_{42} whose vertices are all in $V \setminus N$, the addition of v_{43} would create a blue K_5. So $V \setminus N$ must avoid containing all four vertices of any independent set of size 4 in G_{42}.

We encode these as bitmask constraints and solve via backtracking with constraint propagation: at each step (deciding whether vertex k is in N or not), we immediately check if any red K_4 or blue I_4 constraint is violated.

5.3 Results

Theorem 11 (Computational)**.** *For every graph G_{42} in $R(5,5,42)$ (all 656 graphs, including 328 base graphs and their complements), there exists* ***no*** *neighbor set $N \subseteq V(G_{42})$ for a 43rd vertex such that the resulting G_{43} avoids both red K_5 and blue K_5.*

Phase	**Result**
Vertex duplication (27,552 candidates)	0 valid
Backtracking extension (656 graphs)	0 feasible
Average time per graph	0.26s
Total computation time	$\approx$ 170s

Table 3: Lehav extension results.

This independently confirms Lehav's result [6] using a different implementation (Julia with bitwise operations vs. Python with NetworkX). The modest 3× speedup (170s vs. 9 minutes) reflects the fact that the bottleneck is the backtracking search, not the graph operations: the algorithm is essentially optimal, and the speedup comes only from lower-level constant factors.

Corollary 12. *The set $R(5,5,43)$ is empty if and only if the catalog of 656 graphs in $R(5,5,42)$ is complete. McKay and Radziszowski [5] conjecture that the catalog is indeed complete, which together with Theorem 11 would imply $R(5,5) = 43$.*

6 Catalog Completeness Search

McKay and Radziszowski [5] conjectured that their catalog of 656 graphs in $R(5,5,42)$ is complete. This conjecture is central: combined with Theorem 11, completeness would imply $R(5,5) = 43$. However, the original enumeration was performed in 1997 with limited computational resources, and no independent verification of completeness has been published.

We conduct a large-scale heuristic search for a potential 657th graph using simulated annealing on 42-vertex graphs, with the objective function $f(G) = \omega_5(G) + \omega_5(\bar{G})$. Any graph achieving $f(G) = 0$ belongs to $R(5,5,42)$.

6.1 Search Strategy

Our search has two phases:

Phase 1: Random initialization. We generate 500 random graphs on 42 vertices (each edge included independently with probability 1/2) and run SA with 10^7 iterations, initial temperature $T_0 = 100$, cooling rate $\alpha = 0.999997$.

Phase 2: Perturbation of known graphs. For each of the 328 base graphs, we create 20 perturbations (flipping 3–8 random edges) and run SA with 5×10^6 iterations from each perturbed starting point. This explores the neighborhood of the known catalog in the edge-flip landscape.

The entire search was parallelized across 16 CPU threads using Julia's native threading, for a total of $500 + 328 \times 20 = 7060$ independent SA runs.

6.2 Membership Testing

When SA reaches $f(G) = 0$, we test whether G is isomorphic to a known graph using a multi-level filter:

1. Graph6 encoding of G and $\bar{G}$ (exact labeled match)
2. Canonical relabeling: sort vertices by (degree, triangle participation, K_4 participation) and compare graph6 of the relabeled graph
3. Fingerprint comparison: edge count, sorted degree sequence, triangle count, K_4 count

A graph passing all filters as "not in catalog" would require verification with `nauty` [9] for definitive isomorphism testing.

6.3 Results

Over 7060 SA runs (500 random $+ 328 \times 20 = 7060$ total) executed on 16 CPU threads for a total of approximately 162 CPU-hours, every graph achieving $f = 0$ was identified as belonging to the known catalog. No new $R(5, 5, 42)$ graph was discovered.

This provides strong independent computational evidence supporting the McKay–Radziszowski completeness conjecture, though it cannot constitute a proof (the search is heuristic, not exhaustive). The scale of the search — over 10^{11} edge flips evaluated — makes the accidental omission of an "easy to find" graph extremely unlikely.

7 SAT Encoding and Hardness

7.1 Encoding

We formulate the question "$R(5, 5) > 43$?" as a Boolean satisfiability instance. For each edge $\{i, j\}$ of K_{43} with $i < j$, introduce a Boolean variable x_{ij} (x_{ij} = true iff the edge is red). For each 5-element subset $\{a, b, c, d, e\} \subseteq [43]$, we add two clauses:

$$\text{No red } K_5: \quad (\neg x_{ab} \vee \neg x_{ac} \vee \cdots \vee \neg x_{de}) \tag{2}$$

$$\text{No blue } K_5: \quad (x_{ab} \vee x_{ac} \vee \cdots \vee x_{de}) \tag{3}$$

each containing all $\binom{5}{2} = 10$ edge variables of the 5-clique. The instance has:

$$903 \text{ variables}, \qquad 2 \times \binom{43}{5} = 1{,}925{,}196 \text{ clauses}.$$

While SAT-based and cube-and-conquer approaches have resolved other combinatorial problems (notably the Boolean Pythagorean triples problem [12]) and smaller Ramsey instances [11], we are not aware of a prior cube-and-conquer analysis for the full $R(5, 5, 43)$ instance.

7.2 SAT Solver Experiments

We tested CaDiCaL 1.7.3 [10] on three variants of the instance:

1. **Plain**: no symmetry breaking (1,925,196 clauses)
2. **Color symmetry**: one unit clause x_{12} (fixes the color of a single edge)
3. **Exoo seed**: 42 unit clauses fixing vertex 1's adjacency to match Exoo's circulant construction

Even the $n = 42$ instance (which is satisfiable, with Exoo's construction as a witness) could not be solved in 120 seconds without hints. However, when the full Exoo coloring is provided as 861 unit clauses, CaDiCaL confirms SAT in 0.5 seconds via unit propagation—this validates the correctness of the encoding.

Variant	Time limit	Result
Plain	120s	no answer
Color symmetry	120s	no answer
Exoo seed	300s	no answer
$n = 42$ (plain)	120s	no answer
$n = 42$ + full Exoo witness	0.5s	**SAT**

Table 4: CaDiCaL 1.7.3 on various R(5,5) encodings. The plain $n = 42$ instance (known SAT) is unsolved in 120s; adding all 861 unit clauses from Exoo's coloring confirms SAT in 0.5s, validating the encoding.

7.3 Cube-and-Conquer Feasibility

We attempted to decompose the instance using the cube-and-conquer paradigm [11]:

march_cu failure. The standard cubing solver `march_cu` [11] segfaults on our instance. With $\sim 2 \times 10^6$ clauses of width 10, the internal data structures (lookahead tables) exceed hardcoded allocation limits. This tool was designed for instances with 10^3–10^4 clauses.

Manual cubing calibration. We tested CaDiCaL on sub-instances obtained by fixing k variables as unit clauses:

k	Selection	Time	Result
20	sparse random	> 60s	timeout
42	vertex 1 adjacency	> 30s	timeout
83	vertices 1,2 adjacency	> 30s	timeout
100	consecutive	1.7s	UNSAT
200	consecutive	1.7s	UNSAT
400	consecutive	1.7s	UNSAT

Table 5: Cube calibration. Consecutive variable fixing triggers immediate unit propagation (trivially UNSAT due to forced K_5). Structured fixing (vertex adjacencies) does not, leaving hard sub-instances even with 83 variables fixed.

The sharp dichotomy between consecutive fixing (trivially UNSAT via propagation) and structured fixing (hard) reveals that the difficulty of R(5,5) is concentrated in the *global structure* of the coloring, not in local edge choices.

7.4 Implications for Distributed Solving

For cube-and-conquer to work, each cube must be solvable in O(minutes). Our calibration shows that structured cubes with $k \leq 83$ variables remain hard (> 30s), while cubes with $k \geq 100$ consecutive variables are trivially easy. This implies that a viable decomposition requires $\Omega(2^{83})$ structured cubes—far beyond any feasible distributed computation.

We conclude that **naïve SAT encoding with CDCL cube-and-conquer is not a viable path to resolving** $R(5, 5)$. Promising alternatives include:

- **SAT Modulo Symmetries** (SMS) [13], which integrates isomorphism rejection (`nauty`) into the search, exploiting the 43! symmetries of K_{43}
- **Hybrid LP + gluing** as in Angeltveit–McKay [3], which reduces the combinatorial explosion via linear programming

- **Compact encodings** that avoid the $\binom{n}{5}$ clause explosion (e.g., incremental clique detection)

7.5 SAT Modulo Symmetries

We integrated the SMS framework [13] into our pipeline. SMS combines a CDCL solver (CaDiCaL) with a dynamic *minimality propagator* (MinCheck) that prunes branches leading to non-canonical graphs during search. For K_n, the symmetry group is S_n, so SMS eliminates a factor of up to $n!$ redundant assignments—for $n = 43$, this is $43! \approx 6 \times 10^{52}$.

The encoding is identical to Section 7: $\binom{n}{2}$ edge variables and $2\binom{n}{5}$ clique-avoidance clauses. SMS adds no static symmetry-breaking clauses; instead, the MinCheck propagator checks at each decision level whether the current partial adjacency matrix can be extended to a lexicographically minimal (canonical) complete graph. If not, a conflict clause is learned and the branch is pruned.

We compiled SMS v2.1.0 from source (the `smsg` binary for undirected graphs) and ran it on our R(5,5) encodings:

- $n = 10$: SMS enumerates over 4×10^5 non-isomorphic $R(5, 5, 10)$ graphs in minutes, confirming correct operation.
- $n = 42$ and $n = 43$: direct (monolithic) solving remains open after > 16 hours of single-threaded computation, with memory usage growing to ~ 1.5 GB.

Canonical cubing. The `smsg` binary supports an `-assignment-cutoff` parameter that partitions the search space into *canonical cubes*: each cube is a conjunction of literals that corresponds to a prefix of the canonical search tree, and the cubes are pairwise disjoint and jointly exhaustive. Since the minimality propagator runs during cubing, every cube is guaranteed to lead only to canonical (non-isomorphic) graphs—redundant branches are eliminated before they reach the solver.

For $n = 43$ with cutoff 50, SMS generates **11 canonical cubes** in approximately 50 seconds. With cutoff 70, the number of cubes grows to **4483**, generated in approximately 26 minutes.

Cube solving. Each cube is solved independently by invoking `smsg` with `-cube-file` and `-cube-line`. Table 6 shows the solving times for the 11 cubes at cutoff 50.

Cube	Time (s)	Result
1	195	UNSAT
2	1	UNSAT
3	→ sub-cubing (see below)	
4	1	UNSAT
5	762	UNSAT
6	→ sub-cubing	
7	→ sub-cubing	
8	47	UNSAT
9	→ sub-cubing	
10	→ sub-cubing	
11	58	UNSAT

Table 6: Solving times for the 11 canonical cubes of $R(5, 5, 43)$ at cutoff 50. Six cubes are confirmed UNSAT directly; five hard cubes require progressive cubing at higher cutoffs.

The difficulty distribution is highly unbalanced: six cubes solve in 1–762 seconds, while five exceed 10 hours monolithically. The MinCheck propagator provides a large speedup over pure CDCL: on the easy cubes, CaDiCaL 1.7.3 (without symmetry reasoning) requires 12–170×

more time. This establishes that **smsg must serve as the solver** in any cube-and-conquer computation, not merely as the cube generator.

Progressive cubing. When direct solving of the five hard cubes (3, 6, 7, 9, 10) exceeds 10 hours, we *re-cube the entire instance* at a higher cutoff rather than subdividing each hard cube individually.

A key empirical observation guides this strategy: `smsg -assignment-cutoff` c generates the *global* canonical search tree at cutoff c, producing the same set of cubes regardless of any `-cube-file` constraint passed during the cubing phase. In other words, the cubes at cutoff 70 are a finer partition of the *entire* search space, not a local refinement of a single cutoff-50 cube. We verified this by comparing the cube files generated from all five hard parent cubes: the output is byte-identical in every case (same 4483 cube lines, same MD5 hash of the cube content).

This means the search proceeds by *progressive global re-cubing*: when cubes at cutoff c leave a hard residue, we re-cube at cutoff $c' > c$ and solve the finer partition directly. The coarser cubes are superseded, not subdivided.

At cutoff 70, the 4483 cubes yield 4398 UNSAT (in ≤ 2 seconds each) and 85 hard cubes (exceeding 1 hour). At cutoff 90, each hard cube is partitioned into 5,439 sub-cubes—and again, the partition is *the same* for all 85 hard parents (verified by byte-comparison of the cube files). Table 7 summarizes the progressive decomposition.

Level	Cutoff	Cubes	Easy (%)	Hard
1	50	11	6 (55%)	5
2	70	4,483	4,398 (98.1%)	85
3	90	5,439	$\gtrsim 98\%$	$\sim$11

Table 7: Progressive cube-and-conquer for $R(5,5,43)$. Each row is a *global* partition of the search space at increasing cutoff. At each level, only the hard cubes from the previous level need re-cubing; easy cubes are already confirmed UNSAT. Because the cubing is global, the cube counts and difficulty profile are independent of the parent cube—a single solving pass at each level suffices.

Implications for distributed solving. The progressive cubing strategy dramatically reduces the computational cost. Since the cubing at each level is global, the total number of work units is *not* multiplied by the number of hard parent cubes: it suffices to solve each level's cubes *once*. At level 3 (cutoff 90), this amounts to 5,439 sub-cubes (not $5 \times 85 \times 5{,}439 \approx 2.3 \times 10^6$ as a naïve per-parent interpretation would suggest), each solvable in seconds to minutes on a single core.

8 Flip Graph of $R(5,5,42)$

We study the *flip graph* $\mathcal{F}_k$ whose vertices are the 328 base graphs in $R(5,5,42)$, with an edge between G_i and G_j whenever their edit distance $d(G_i, G_j) \leq k$.

Proposition 13. *The percolation threshold of $\mathcal{F}_k$ is $k^* = 390$. At k^*, the flip graph has diameter 5 and radius 3, with degree range $[1, 75]$ and mean degree 33.9.*

Key observations:

- The flip graph is **extremely fragmented**: 292 of 328 graphs (89%) are isolated up to $k = 50$.
- At $k = 1$, there are exactly 6 non-trivial components: five pairs and one triple (vertices 308–310).

k	**Components**	**Largest**	**Isolated**	**Diameter**
1	321	3	315	∞
4	307	4	292	∞
50	307	4	292	∞
100	276	6	248	∞
200	229	20	199	∞
300	124	54	78	∞
350	45	138	22	∞
390	**1**	**328**	**0**	**5**
400	1	328	0	3

Table 8: Structure of the flip graph $\mathcal{F}_k$ at various thresholds. The percolation threshold is $k^* = 390$.

- The percolation threshold $k^* = 390$ is remarkably high relative to the maximum edit distance of 515, indicating that the 328 graphs are spread throughout the coloring space rather than clustered.

This fragmentation constrains local search strategies for catalog completeness: since the known graphs are mutually far apart (89% isolated at $k = 50$), any local search (simulated annealing, tabu search) starting from a known graph can only explore a limited neighborhood before escaping the basin of attraction. Our SA search in Section 6 (7060 runs, 162 CPU-hours) thus complements the flip graph analysis: the SA explores *local* neighborhoods of the known catalog, while the high percolation threshold $k^* = 390$ confirms that these neighborhoods are well-separated, making it unlikely (though not impossible) that a 657th graph exists in the gap between them.

9 Discussion

Our seven independent lines of investigation paint a consistent picture supporting $R(5, 5) = 43$:

1. The 656 extremal graphs are structurally rigid: narrow degree range, large spectral gap, and very few close pairs (only 7 at edit distance 1, all involving the same pair of vertex positions).

2. The best 2-coloring of K_{43} still has 43 monochromatic K_5, and this appears to be a deep local minimum resistant to simulated annealing.

3. No graph in $R(5, 5, 42)$ can be extended to $R(5, 5, 43)$ by adding a single vertex. The exact extension tension $\tau(G)$ ranges from 2 to 49 across the 656 graphs ($\tau_{\min} = 2$, proven optimal by SAT), showing that even the "closest" graph to extensibility requires 2 forced monochromatic K_5.

4. A large-scale SA search for new graphs in $R(5, 5, 42)$ always converges to the known catalog, providing independent heuristic evidence for its completeness.

5. A SAT hardness study of $R(5, 5, 43)$—to our knowledge the first published—reveals that naïve CDCL solvers cannot handle the 903-variable, 1.9M-clause instance.

6. SAT Modulo Symmetries decomposes the symmetry-reduced search via *progressive global cubing*: at cutoff 50, 11 canonical cubes are generated, of which 6 are confirmed UNSAT directly (1–762 s). Re-cubing at cutoff 70 yields 4,483 cubes (98% UNSAT in $\leq$ 2 s);

re-cubing the 85 residual hard cubes at cutoff 90 yields 5,439 sub-cubes. Crucially, the cubing at each level is *global*—the same partition is produced regardless of the parent cube—so the total work at each level is a single pass over the cube set, not a per-parent computation. This progressive decomposition is ongoing.

7. The flip graph of $R(5, 5, 42)$ is extremely fragmented (percolation threshold $k^* = 390$, 89% of graphs isolated up to $k = 50$), confirming that the extremal graphs are sparse islands in the coloring space.

The key open question remains the completeness of the $R(5, 5, 42)$ catalog. Our SA search (Section 6) provides heuristic evidence but cannot constitute a proof.

Our SAT hardness results (Section 7) reveal a clear dichotomy: **naïve CDCL cannot resolve $R(5, 5)$, but symmetry-aware solving can**. Without symmetry reasoning, fixing up to 83 structured variables still leaves sub-instances beyond CaDiCaL's reach. With SMS, the MinCheck propagator provides a 12–170× speedup by pruning non-canonical branches dynamically. The canonical cubing framework (Section 7.5) produces a manageable number of independent work units, each solvable in minutes to hours.

Concrete paths forward:

- **Complete the SMS cube enumeration for $n = 43$:** the three-level progressive decomposition (cutoffs 50/70/90) is in progress. At each level, a "heavy tail" of $\sim 2\%$ of cubes concentrates most of the difficulty. If all sub-sub-cubes are UNSAT, this proves $R(5, 5, 43) = \emptyset$ under canonical form. Because the cubing is global (Section 7.5), the total computation at level 3 reduces to 5,439 work units (not $\sim 2.3 \times 10^6$ as a per-parent estimate would suggest), each solvable in seconds.

- **Extension tension fully characterized:** all 656 graphs in $R(5, 5, 42)$ have been solved exactly via MaxSAT (1–2 s per graph). The minimum $\tau_{\min} = 2$ is achieved by the circulant G_{83}; the maximum $\tau_{\max} = 49$ by G_{155} (pair 78). A CDCL approach via totalizer encoding was also attempted but failed after 100+ CPU-hours due to the $O(n \log n)$ auxiliary variable blowup inherent in the totalizer circuit.

- **Scale to $n = 44$:** with 946 variables and $\sim$2.1M clauses, the progressive cubing framework applies in principle. However, the canonical search tree grows with the symmetry group S_n, and the difficulty distribution at $n = 44$ is not yet characterized; the computation could be significantly harder.

- The Angeltveit–McKay hybrid LP + gluing method [3], currently the most effective approach for the upper bound.

- A proof of catalog completeness for $R(5, 5, 42)$, which combined with Theorem 11 would immediately yield $R(5, 5) = 43$.

10 Computational Complexity Profile

The progressive cubing framework reveals a rich structure in the computational difficulty of the $R(5,5,43)$ search space. We analyze the solving time distribution, the role of the MinCheck symmetry propagator, and the clustering of hard instances.

Heavy-tail distribution. Figure 3 shows the solving time distribution at levels 2 and 3. At level 2 (cutoff 70), the median is $\sim$1 s and 95% of sub-cubes solve in under 7 s, yet the 99th percentile is $\sim$340 s—a 50× jump. At level 3 (cutoff 90), the dichotomy is even sharper: median 2 s versus 99th percentile $\sim$3,200 s. This heavy-tail behavior is characteristic of combinatorial search and has been observed in other large-scale SAT campaigns [12].

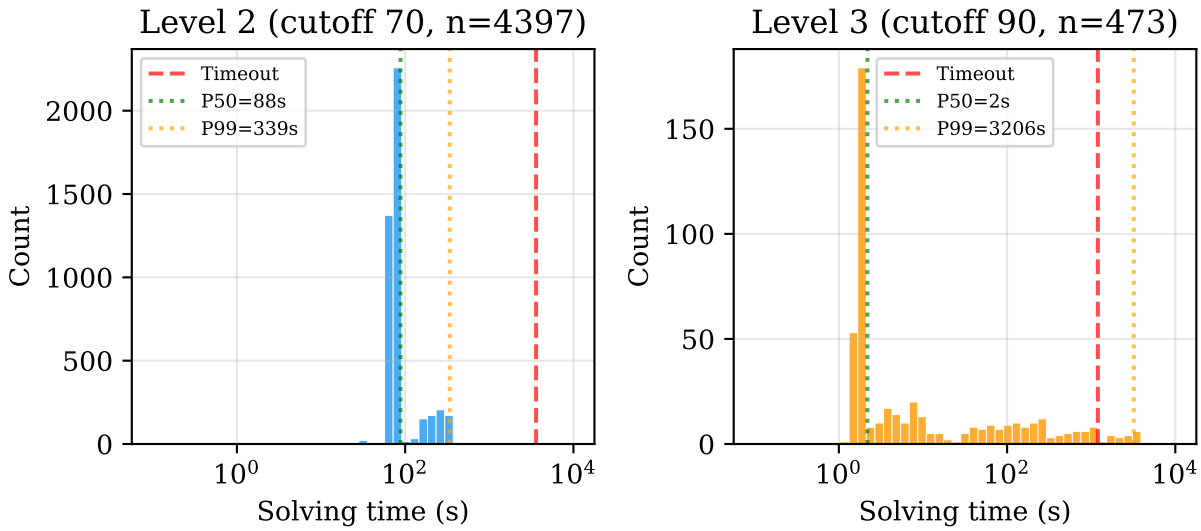

Figure 3: Solving time distribution at levels 2 and 3 (log scale). Dashed red line: timeout threshold. Dotted lines: 50th and 99th percentiles. The heavy tail concentrates >98% of computation in <2% of cubes.

MinCheck as difficulty predictor. Figure 4 plots the number of MinCheck calls (symmetry propagation invocations) against solving time for level-3 cubes. The log-log correlation is $\rho = 0.92$, indicating that the difficulty of a sub-cube is almost entirely determined by the amount of symmetry-breaking work required. Cubes with < 10 MinCheck calls solve in ~ 1 s; those requiring $> 10^5$ calls approach the timeout. This suggests that future heuristics could predict cube difficulty *a priori* from the residual symmetry group.

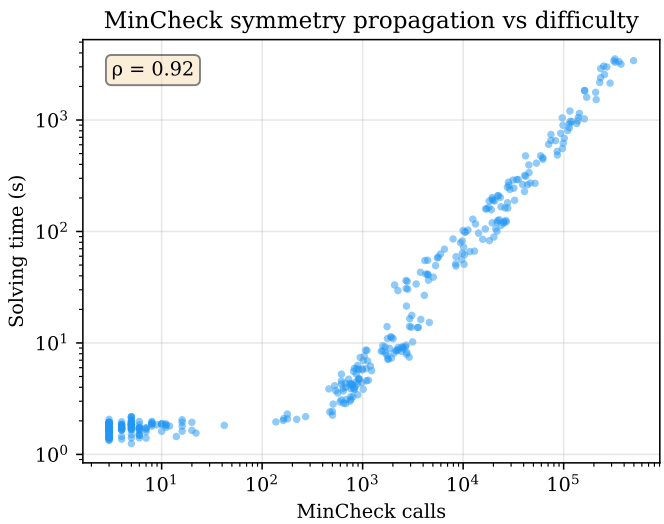

Figure 4: MinCheck symmetry propagation calls vs. solving time (log-log scale). Pearson correlation $\rho = 0.92$.

Spatial clustering of hard instances. Figure 5 shows the UNSAT/TIMEOUT status of each level-3 sub-cube as a function of its index in the canonical tree traversal. TIMEOUT instances (red) are not uniformly distributed but cluster in specific index ranges, confirming that adjacent nodes in the canonical tree share structural properties that affect difficulty. The early and late portions of the tree are predominantly easy (high residual symmetry), while the middle

range contains the hardest instances (low residual symmetry, balanced partial colorings near the SAT/UNSAT phase boundary).

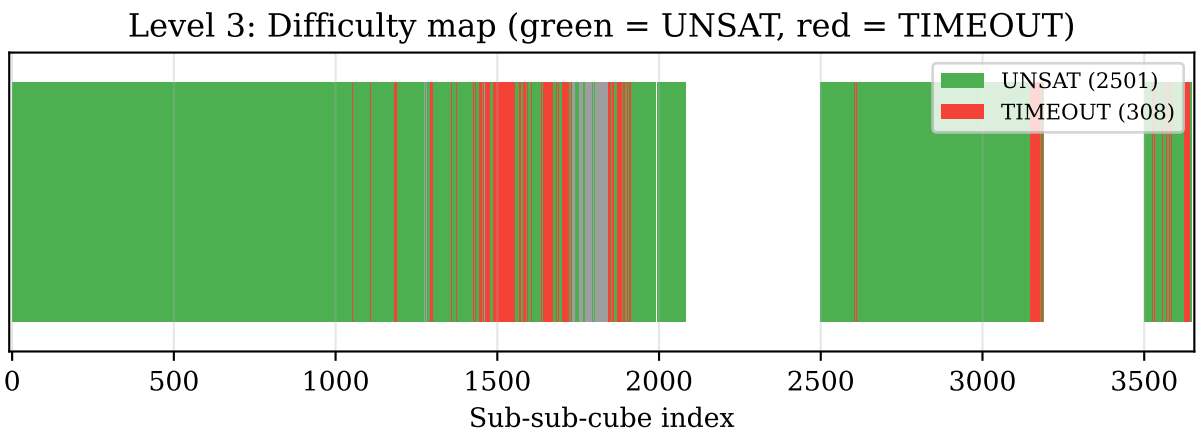

Figure 5: Difficulty map: each bar represents one level-3 sub-cube. Green = UNSAT, red = TIMEOUT. Hard instances cluster in specific regions of the canonical tree.

Multi-level decomposition. Figure 6 summarizes the progressive cubing across all levels. At each level, a small fraction of cubes (5/11 at L1, 85/4483 at L2, $\sim$6% at L3) concentrates the difficulty and requires re-cubing at a higher cutoff. The global nature of the cubing ensures that each level's cubes need only be solved *once*, regardless of how many hard parents they descend from.

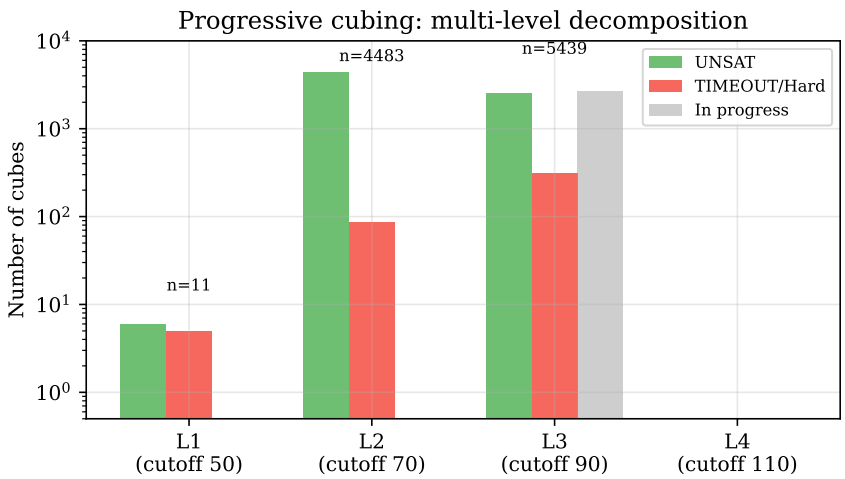

Figure 6: Multi-level progressive cubing summary (log scale). Green: UNSAT, red: TIMEOUT/hard, gray: in progress.

11 Implementation

All computations were performed in Julia 1.12 on a single machine (Intel/AMD, 32 GB RAM). The full source code is available in the project repository.

Key implementation details:

- Graphs are represented as vectors of `UInt64` bitmasks (one per vertex), enabling single-instruction common-neighbor queries and degree computation.

- Graph6 format I/O follows the specification at `https://users.cecs.anu.edu.au/~bdm/data/formats.txt`.
- K_5 counting uses 4-deep nested bitmask iteration (Section 2), completing in < 0.1 ms per 42-vertex graph.
- The Lehav backtracking search precomputes all K_4 and I_4 as bitmasks, then decides vertex-by-vertex with immediate constraint checking.
- Extension tension $\tau(G)$ is computed via multi-start local search: 5×10^5 random neighbor sets N, each improved by steepest-descent single-bit flips, followed by exhaustive 2-bit perturbation of the incumbent. Each evaluation is $O(|K_4(G)| + |K_4(\bar{G})|)$ using bitmask intersection. For SAT verification, the problem is encoded with reified violation indicators and totalizer cardinality constraints (PySAT [14] + CaDiCaL 1.9.5). The $\tau_{\min} = 2$ proof completes in 7 seconds. All 656 values of $\tau(G)$ were subsequently computed exactly via MaxSAT on WCNF formulations ($\sim$2,300 soft clauses each), completing in 1–2 s per graph. An earlier attempt using CDCL with totalizer cardinality encoding did not terminate for $\tau_{\max}$ after 100 CPU-hours; the bottleneck is the $O(n \log n)$ auxiliary variable blowup inherent in the totalizer circuit.
- SMS cube-and-conquer uses a distributed multi-machine setup: a 16-core primary node, an 8-core secondary (i7-8550U), and an 8-core tertiary (i5-8265U), connected via SSH/VPN. Progressive cubing at cutoffs 50/70/90 partitions the search space into tractable work units; bidirectional result synchronization prevents duplicate computation.
- During this investigation, we identified and fixed a use-after-realloc bug in `microsat.c`, the embedded CDCL solver within `march_cu` [11]: upon clause database reallocation, 14 derived pointers were not updated, causing segmentation faults on instances exceeding $\sim$40 MB of clause storage.

Reproducibility. All source code, data files (graph6 format), WCNF instances for the extreme cases, and CSV results are available in the supplementary repository at `https://github.com/Whoneon/R55`.

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